

AI Engineer Roadmap — 6 Months

Python → ML → Deep Learning → LLMs → Deploy → Job Ready

MONTH 01

Weeks 1 – 4

Python & Math Foundations

- Python — loops, functions, OOP, NumPy, Pandas
- Linear Algebra — vectors, matrices, dot products
- Statistics & Probability — distributions, variance
- Calculus — derivatives & gradients for ML

[freeCodeCamp](#) • [Khan Academy](#) • [3Blue1Brown](#)

01

MONTH 02

Weeks 5 – 8

Data Analysis & Visualization

- Exploratory Data Analysis (EDA) with Pandas
- Data cleaning — missing values, encoding, outliers
- Visualizations — Matplotlib, Seaborn, Plotly
- Feature engineering & dataset preparation

[Kaggle Learn](#) • [Towards Data Science](#)

02

MONTH 03

Weeks 9 – 12

Machine Learning

- Supervised learning — regression, classification, trees
- Unsupervised learning — K-Means, PCA, DBSCAN
- Scikit-learn — pipelines, cross-validation, tuning
- Evaluation metrics — precision, recall, ROC, F1

[Andrew Ng ML Course](#) • [Hands-On ML \(Geron\)](#)

03

MONTH 04

Weeks 13 – 17

Deep Learning & Neural Networks

- Neural networks — layers, activations, backprop
- PyTorch — tensors, custom models, training loops
- CNNs for Computer Vision — ResNet, transfer learning
- RNNs & LSTMs for sequential data

[fast.ai](#) • [deeplearning.ai](#) • [CS231n Stanford](#)

04

MONTH 05

Weeks 18 – 22

NLP, Transformers & LLMs

- NLP — tokenization, embeddings, attention mechanism
- Transformers & BERT via Hugging Face library
- LLMs — GPT architecture, fine-tuning with LoRA
- RAG systems — LangChain + vector DBs (Chroma/FAISS)

[Hugging Face Course](#) • [LangChain Docs](#)

05

MONTH 06

Weeks 23 – 26

Deploy, MLOps & Job Ready

- Model serving — FastAPI, Streamlit, HuggingFace Spaces
- MLOps — Docker, MLflow, Weights & Biases (W&B)
- Build 3 projects: chatbot, image classifier, RAG app
- Polish GitHub + resume + LinkedIn — start applying!

[Made With ML](#) • [Full Stack Deep Learning](#)

06